Jaivant Kosaraju (231IT028) Paluvadi Dinesh Manideep (231IT047) Paluru Pavan Kumar (231IT046)

Department of Information Technology
National Institute of Technology Karnataka, Surathkal

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- **Reference Architecture:** W. Dev, S. Bera, and A. Tiwari, *"Control-Plane Load Balancing and Auto-Scaling in 5G and Beyond Networks"*, IEEE ICOIN 2025.
- **Core Deployment:** Ingress load balancer stationed between gNodeBs and AMFs to shield the core plane.
- **Traffic Routing:** Static round-robin distribution of N2 signaling messages across active AMF replicas.
- **Reactive Threshold Policy:** Triggers scale-out when active load ratio exceeds capacity:

$$\frac{B}{A \cdot \alpha} > 1.0$$

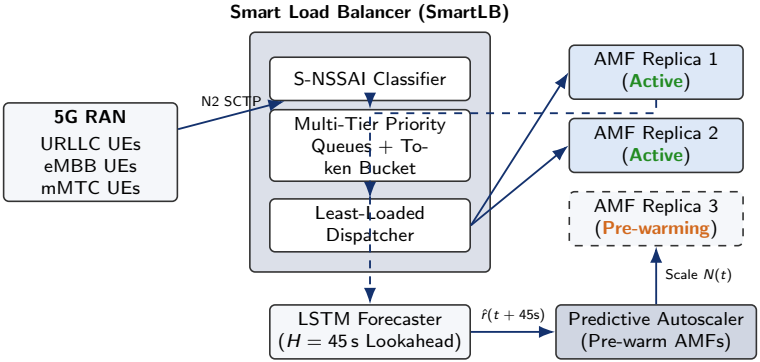
- **Proactive Threshold Policy:** Adds an AMF early when the ratio crosses a safety threshold γ (typically $\gamma = 0.8$):

$$\frac{B}{A \cdot \alpha} \geq \gamma$$

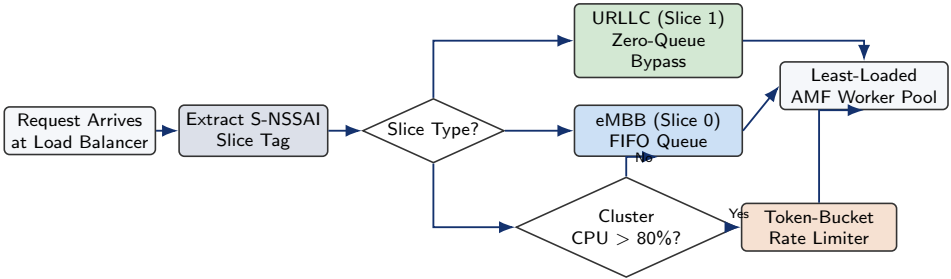
- **Container Startup Lag ($T_{\mathbf{boot}} \approx 40\,\mathbf{s}$):** AMF initialization takes 40 seconds, during which requests accumulate in queues.
- **Reactive Bottleneck:** Autoscaling decisions fire *after* traffic surges hit, causing severe overload windows (54 s).
- **Slice-Agnostic Single FIFO:** Treats all 3GPP traffic classes identically with no priority differentiation.
- **Head-of-Line Blocking:** Massive mMTC IoT registration bursts stall time-critical URLLC signaling messages.
- **SLA Violations:** URLLC tail latency reaches 338.7 ms under surge, failing the 3GPP TS 22.261 5.0 ms SLA.

- **Objective 1: Slice-Aware Priority Queuing (O1)**
  - Classify incoming messages by 3GPP S-NSSAI slice type (URLLC, eMBB, mMTC).
  - Provide zero-queue fast-path bypass for time-critical URLLC signaling.
  - Isolate delay-tolerant mMTC surges using token-bucket rate limiting above 80% CPU.

- **Objective 2: LSTM Predictive Auto-Scaling (O2)**
  - Forecast incoming aggregate request rates $H = 45\,\text{s}$ into the future ($H \geq T_{\text{boot}}$).
  - Pre-warm and initialize AMF container replicas before traffic surges arrive.
  - Completely eliminate cold-start queue accumulation and cut idle resource waste.


Smart Load Balancer (SmartLB)
5G RAN
URLLC UEs
eMBB UEs
mMTC UEs
N2 SCTP
S-NSSAI Classifier
Multi-Tier Priority Queues + Token Bucket
Least-Loaded Dispatcher
AMF Replica 1 (Active)
AMF Replica 2 (Active)
AMF Replica 3 (Pre-warming)
Scale $N(t)$
LSTM Forecaster ($H = 45$ s Lookahead)
$\hat{r}(t + 45\text{s})$
Predictive Autoscaler (Pre-warm AMFs)

- **URLLC (Slice 1 – Mission-Critical):**
  - Strict priority zero-queue bypass directly to least-loaded active AMF instance.
  - 10–15% capacity headroom permanently reserved across AMF instances.
  - Strict latency bound guaranteed under 5.0 ms.
- **eMBB (Slice 0 – Standard Mobile Broadband):**
  - Standard FIFO scheduling queue with round-robin dispatching.
  - Typical latency target between 10–25 ms.
- **mMTC (Slice 2 – Delay-Tolerant IoT):**
  - Buffered queue with dynamic token-bucket rate limiting when AMF CPU $> 80\%$.
  - Starvation protection aging mechanism promotes packets waiting over 5.0 s.



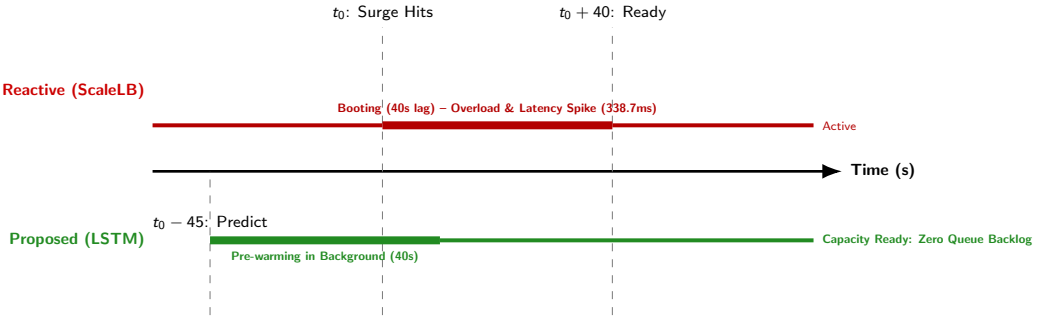
- **URLLC Priority:** Immediate dispatch; bypasses all intermediate queue buffers.
- **mMTC Protection:** Heavy IoT traffic absorbed and regulated without starving standard users.

- **Input Observation Window:** Past 60 seconds ($W = 60\,\text{s}$) of aggregate arrival rates.
- **Lookahead Prediction Horizon:** Forecasts load $H = 45\,\text{s}$ ahead ($H \geq T_{\text{boot}} = 40\,\text{s}$).
- **Fleet Capacity Dimensioning Formula:**

$$N(t) = \left\lceil \frac{\hat{r}(t+H) \cdot (1+m)}{\mu \cdot \gamma_{\text{target}}} \right\rceil$$

  - $\hat{r}(t+H)$: LSTM predicted request rate at future horizon.
  - $\mu = 100\,\text{req/s}$: Maximum service throughput capacity of one AMF instance.
  - $\gamma_{\text{target}} = 0.80$: Target utilization cap preserving 20% headroom.
  - $m = 0.15$: Headroom safety buffer guarding against forecast uncertainty.

- **Hysteresis Stabilization:** 30-second cooldown prevents container flapping.



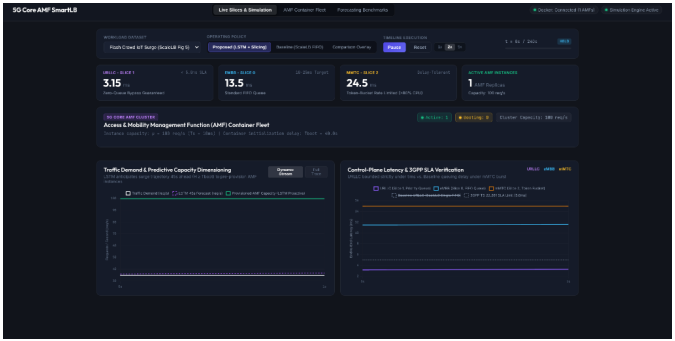
- **Reactive ScaleLB Failure:** Triggers only after surge hits; suffers 54 s overload and 338.7 ms tail latency.
- **Predictive Pre-Warming:** New instances boot 45 s early; full capacity available before traffic arrives.

- **Docker Microservices Pipeline:**
  - `smart-loadbalancer`: Reverse proxy with S-NSSAI parser, 3 queues, and token bucket.
  - `amf-worker`: Scalable container replicas simulating 5G registration procedures.
  - `traffic-replayer`: High-throughput workload generator reproducing 3GPP traces.
- **SimPy Discrete-Event Simulation Engine:**
  - Replays 240-second flash-crowd surges with deterministic sub-millisecond precision.
  - Evaluates queuing delay, token replenishments, and 40-second container startup lags.
- **Web Operations Platform:**
  - Real-time dashboard with dynamic capacity charts, per-slice KPIs, and container fleet view.

Configuration	Cold-Start Lag (s)	URLLC p99 Latency	mMTC p99 Latency	URLLC SLA Compliance	AMF-sec (Cost)
ScaleLB Reactive (Baseline)	54 s	338.7 ms	339.3 ms	**FAIL (25.8% viol.)**	**647**
ScaleLB Proactive ($\gamma = 0.8$)	55 s	334.1 ms	336.5 ms	**FAIL (25.8% viol.)**	649
Warm Standby (1 spare)	**0 s**	32.9 ms	32.7 ms	**PASS (0.0% viol.)**	1024
O1: Slice Priority Queue Only	54 s	11.1 ms	508.6 ms	**PASS (0.0% viol.)**	**647**
O2: LSTM Prediction Only	40 s	333.1 ms	336.5 ms	**FAIL (18.3% viol.)**	807
Proposed Full System (O1 + O2)	**0 s**	**3.15 ms**	**499.9 ms**	**PASS (0.0% viol.)**	**807**

- **Baseline Failure:** ScaleLB suffers 338.7 ms tail latency and 25.8% SLA failure.
- **Proposed Success:** Achieves 3.15 ms URLLC latency with zero cold-start delay.

- **Complete Cold-Start Elimination:**
  - Lookahead prediction ($H = 45\,\text{s}$) brings 4 AMF replicas online before surge arrival.
  - Reduces cold-start overload lag from 54 s down to **0 s**.
- **URLLC 3GPP SLA Compliance:**
  - Priority bypass drops 99th-percentile latency from 338.7 ms to **3.15 ms**.
  - 100% compliance with 3GPP TS 22.261 sub-5 ms latency mandate.
- **21.2% Infrastructure Cost Reduction:**
  - Consumes only 807 AMF-seconds compared to 1024 AMF-seconds under Warm Standby.
  - Avoids wasting CPU and memory on permanently running idle reserve containers.



- **Real-Time Workload Tracking:** Visualizes incoming request rates and 45s predictive trajectory.
- **Fleet Auto-Scaling:** Live display of active, booting, and decommissioned AMF container instances.
- **Latency SLAs by Slice:** Real-time verification of sub-5 ms URLLC compliance alongside buffered mMTC delay.

- **Project Summary:**
  - Integrated 3GPP S-NSSAI priority queuing with LSTM predictive autoscaling.
  - Fully eliminated the 40-second container cold-start bottleneck.
  - Guaranteed 3.15 ms URLLC latency with zero SLA violations during peak surge.

- **Future Work:**
  - **Multi-Region Federation:** Distribute AMF autoscaling across edge and regional data centers.
  - **Kernel-Bypass Ingress:** Implement eBPF/XDP dispatching for line-rate 100 Gbps processing.

- **[1]** W. Dev, S. Bera, and A. Tiwari, *"Control-plane load balancing and auto-scaling in 5G and beyond networks"*, in *Proc. IEEE ICOIN*, 2025, pp. 83–87.
- **[2]** 3GPP, *"Service requirements for the 5G system"*, 3GPP TS 22.261, Release 18, 2023.
- **[3]** Open5GS Project, *"Open source 5G core network implementation"*, `https://open5gs.org`.
- **[4]** S. Hochreiter and J. Schmidhuber, *"Long short-term memory"*, *Neural Computation*, vol. 9, no. 8, pp. 1735–1780, 1997.